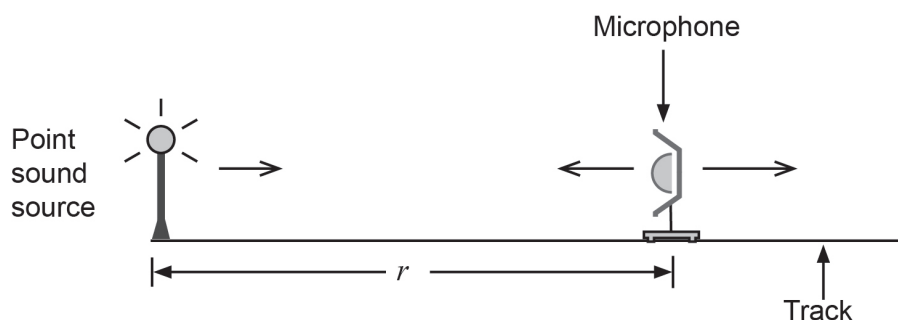


## Question 14

(17 marks)

Sound intensity is defined as sound power per unit area,  $\text{Intensity} = \frac{\text{Power}}{\text{Area}}$  ( $I = \frac{P}{A}$ ). Hence, the further away from a source, the more area (spherical area =  $4\pi r^2$ ) the sound is spread across, and the less intense a sound appears to a person.

A physics student was asked to verify the formula  $I = P/A$ . The experimental set-up was as follows.

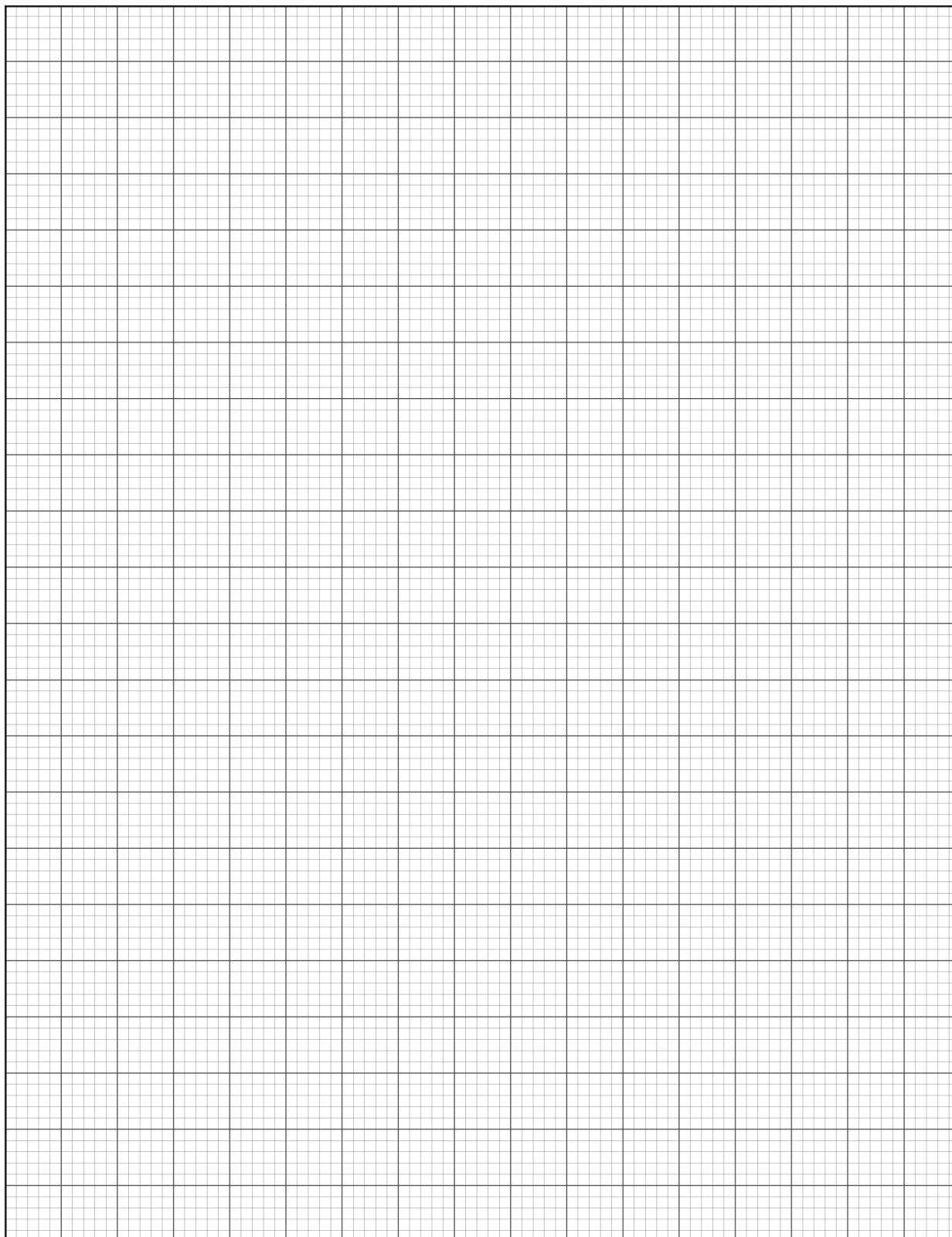


The student used a microphone to monitor the level of the intensity of the sound at various distances. As the student moved the microphone along the track, the values were recorded in a table as follows.

| Intensity, $I$ ( $\text{W m}^{-2}$ ) | Distance from speaker, $r$ (m) | $\frac{1}{r^2}$ ( $\text{m}^{-2}$ ) |
|--------------------------------------|--------------------------------|-------------------------------------|
| $48 \pm 1$                           | 0.5                            | 4.0                                 |
| $10 \pm 1$                           | 1.0                            | 1.0                                 |
| $5 \pm 1$                            | 1.5                            |                                     |
| $3 \pm 1$                            | 2.0                            | 0.25                                |
| $2 \pm 1$                            | 2.5                            |                                     |

- (a) Complete the last column in the table above with values expressed to **two** significant figures. (3 marks)
- (b) Use the data from the table to plot a straight line graph (including error bars) on the grid provided, demonstrating the relationship between the intensity ( $\text{W m}^{-2}$ ) and  $\frac{1}{r^2}$  ( $\text{m}^{-2}$ ). (5 marks)

DO NOT WRITE IN THIS AREA AS IT WILL BE CUT OFF



If you wish to make a second attempt at this item, the grid is repeated at the end of this Question/Answer Booklet. Indicate clearly on this page if you have used the second grid and cancel the working on this page.

**See next page**

**Question 14** (continued)

- (c) Determine the gradient of your line of best fit. Show **all** workings, and include units in your answer. (3 marks)
- (d) Use your graph to determine the intensity at a distance of 0.70 m from the source. Show **all** workings, and express your answer using appropriate significant figures. (3 marks)
- (e) Using the answer from (d), calculate the power output of the source. Show appropriate units. (2 marks)
- (f) Using your answers to (c) and (e), did the physics student verify the formula  $I = P/A$ ? (1 mark)

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